Assignment 10: Data Scraping

Student Name

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:

- Load the packages `tidyverse`, `rvest`, and any others you end up using.
- Check your working directory

```
#1

library(tidyverse)
library(rvest)
library(here)
library(lubridate)

here()
```

```
## [1] "/Users/kaitlynkukula/EDE_Fall2025"
```

```
kak_theme <- theme_classic() +
  theme(plot.title = element_text(size = 11, hjust = 0.5, face = "bold"), #Adjust title
        axis.text.x = element_text(size = 9), # Adjust x-axis values
        axis.text.y = element_text(size = 9), # Adjust y-axis values
        axis.title.x = element_text(size = 10.5, face = "bold"), # Adjust x-axis title
        axis.title.y = element_text(size = 10.5, face = "bold"), # Adjust y-axis title
        legend.position = "bottom", # Define legend position
        legend.text = element_text(size = 9), # Define legend entry sizes
        legend.title = element_text(size = 9)) # Define legend name sizes
set_theme(kak_theme)
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham's 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to https://www.ncwater.org/WUDC/app/LWSP/search.php
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2

lwsp_link <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024')
```

3. The data we want to collect are listed below:

- From the "1. System Information" section:
- Water system name
- PWSID
- Ownership
- From the "3. Water Supply Sources" section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

> HINT: The first value should be "Durham", the second "03-32-010", the third "Municipality", and the last should be a vector of 12 numeric values (represented as strings)".

```
#3
water_system <- lwsp_link %>% html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>% html_text()
pwsid <- lwsp_link %>% html_nodes('td tr:nth-child(1) td:nth-child(5)') %>% html_text()
owner <- lwsp_link %>% html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>% html_text()
mgd_monthly <- lwsp_link %>% html_nodes('th~ td+ td') %>% html_text()
```

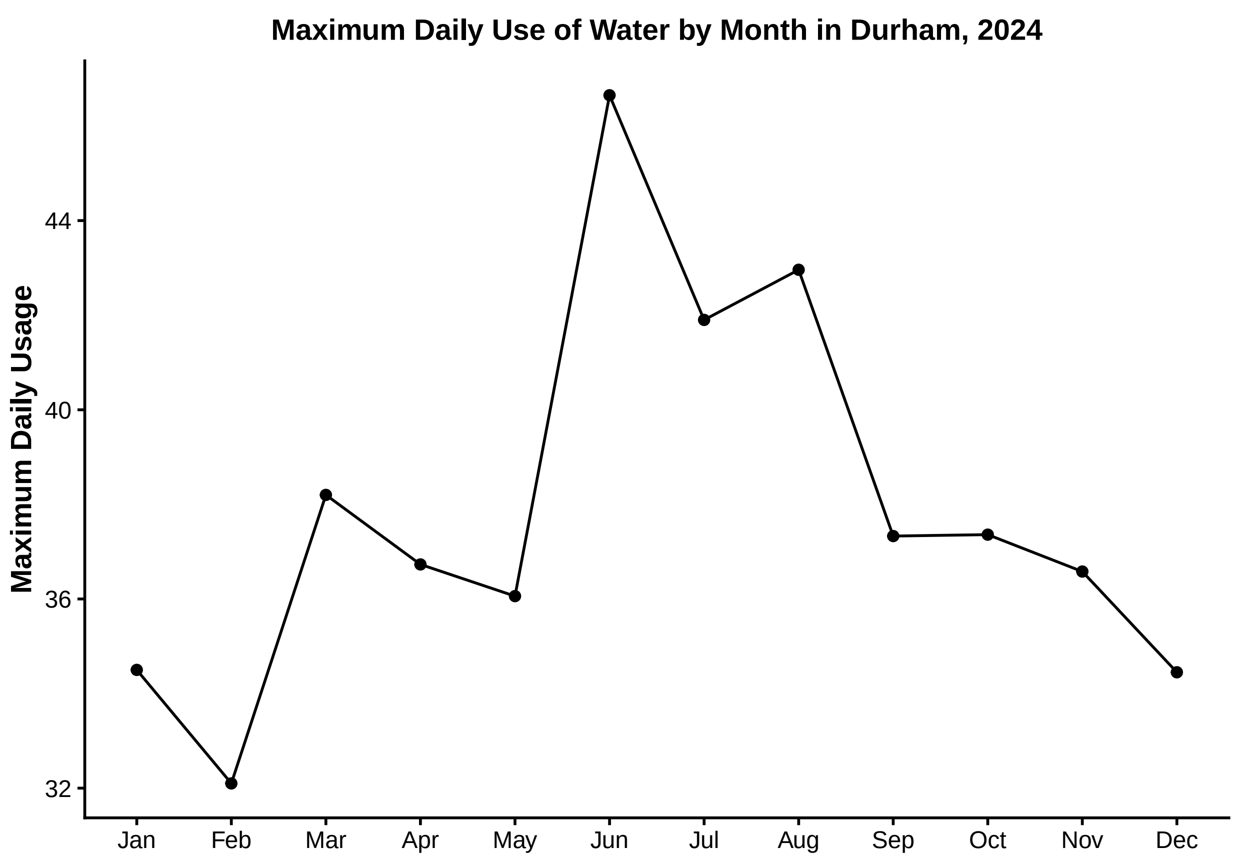
4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

   TIP: Use `rep()` to repeat a value when creating a dataframe.

   NOTE: It's likely you won't be able to scrape the monthly widthrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc…

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4

month = c("01", "05", "09",
            "02", "06", "10",
            "03", "07", "11",
            "04", "08", "12")
year = rep("2024", 12)

water_dataframe <- data.frame(
  date = dmy(paste0("01-",month,"-",year)),
  water_system = water_system,
  pwsid = pwsid,
  owner = owner,
  mgd_monthly = as.numeric(mgd_monthly))

#5
ggplot(water_dataframe, aes(x = month(date), y = mgd_monthly)) +
  geom_point() +
  geom_line() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(x = NULL,
       y = "Maximum Daily Usage",
       title = "Maximum Daily Use of Water by Month in Durham, 2024")
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a

function with two inputs - “PWSID” and “year” - that:

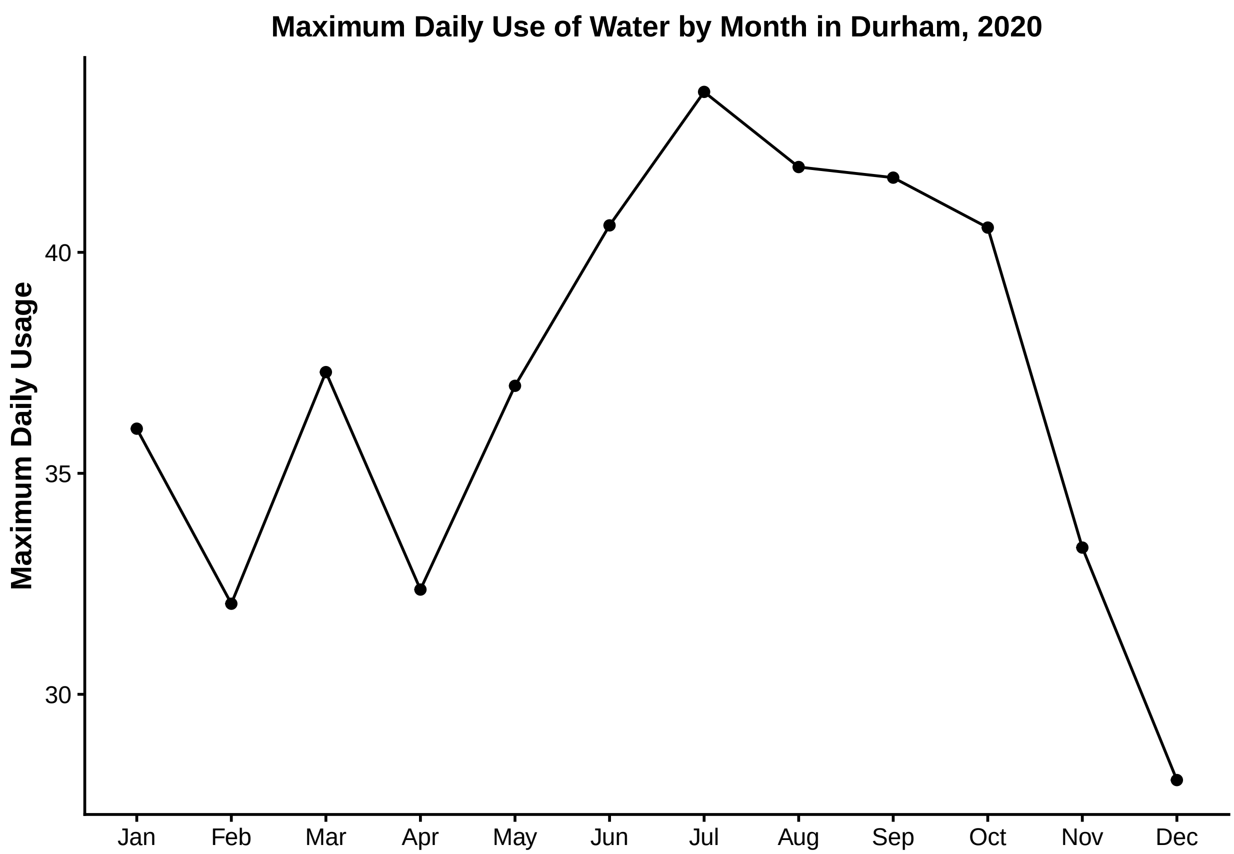
- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```r
#6.
pwsid_input <- 03-32-010
year_input <- 2024

lwsp_function <- function(pwsid_input, year_input) {
# Get the URL
   lwsp_fcn_url <- paste0('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=',
           pwsid_input,'&year=',year_input)

# Fetch the website
    lwsp_fcn_website <- read_html(lwsp_fcn_url)

# Scrape the data
    water_system_fcn <- lwsp_fcn_website %>% html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %
    pwsid_fcn <- lwsp_fcn_website %>% html_nodes('td tr:nth-child(1) td:nth-child(5)') %>% html_text()
    owner_fcn <- lwsp_fcn_website %>% html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>% html_
    mgd_monthly_fcn <- lwsp_fcn_website %>% html_nodes('th~ td+ td') %>% html_text()

# Construct a data frame

    month = c("01", "05", "09",
            "02", "06", "10",
            "03", "07", "11",
            "04", "08", "12")
    year = rep(year_input, 12)

    water_dataframe_fcn <- data.frame(
      date = dmy(paste0("01-",month,"-",year)),
      water_system_fcn = water_system_fcn,
      pswid_fcn = pwsid_input,
      owner_fcn = owner_fcn,
      mgd_monthly_fcn = as.numeric(mgd_monthly_fcn))

# Return the dataframe
    return(water_dataframe_fcn)
}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID=‘03-32-010’) for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```r
#7
water_durham_2020 <- lwsp_function('03-32-010', 2020)

ggplot(water_durham_2020, aes(x = month(date), y = mgd_monthly_fcn)) +
```

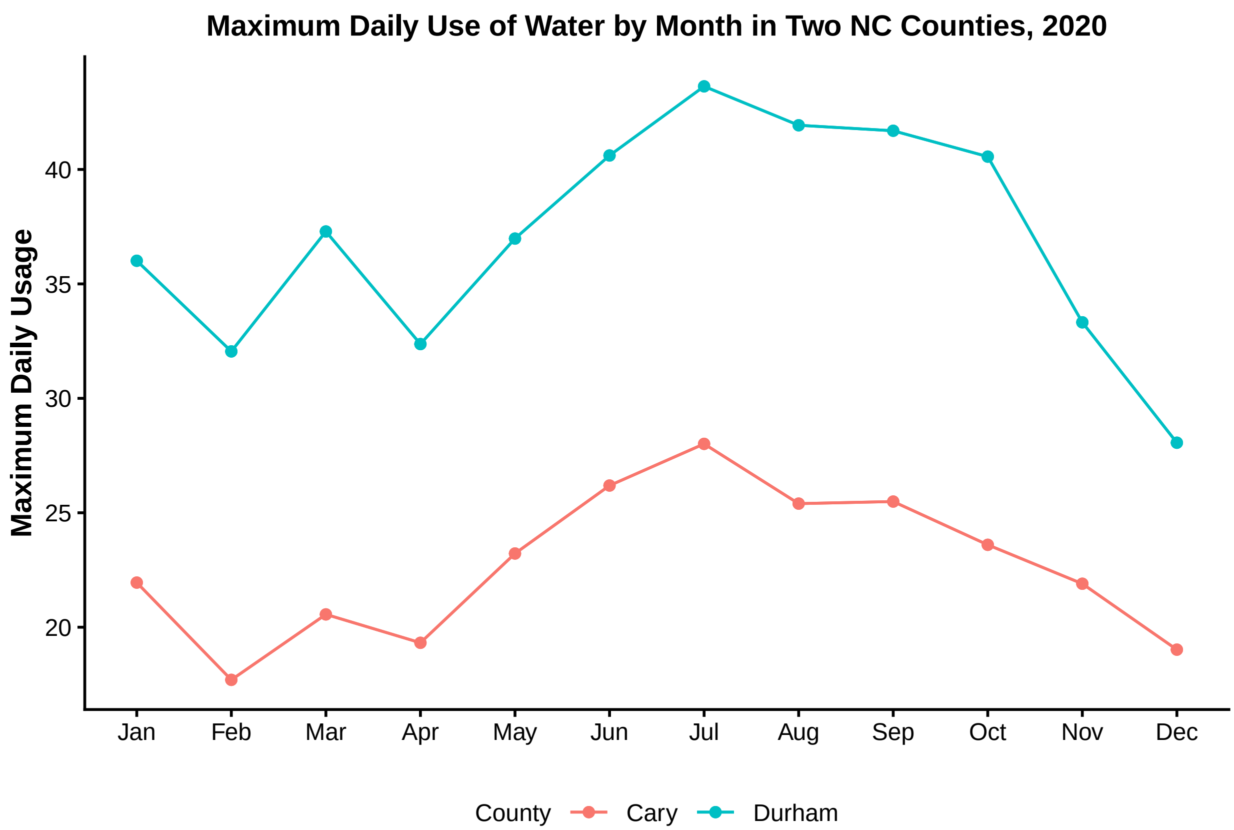
```
  geom_point() +
  geom_line() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(x = NULL,
       y = "Maximum Daily Usage",
       title = "Maximum Daily Use of Water by Month in Durham, 2020")
```



8. Use the function above to extract data for Cary (PWSID = '03-32-010') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

```
#8
# Get Cary 2020 data
water_cary_2020 <- lwsp_function('03-92-020', 2020)

# Combine Cary and Durham data
water_nc_2020 <- rbind(water_durham_2020, water_cary_2020)

# Plot Cary and Durham data
ggplot(water_nc_2020) +
  geom_point(aes(x = month(date), y = mgd_monthly_fcn, color = water_system_fcn)) +
  geom_point(aes(x = month(date), y = mgd_monthly_fcn, color = water_system_fcn)) +
  geom_line(aes(x = month(date), y = mgd_monthly_fcn, color = water_system_fcn)) +
  geom_line(aes(x = month(date), y = mgd_monthly_fcn, color = water_system_fcn)) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
```

```r
  labs(x = NULL, y = "Maximum Daily Usage",
       title = "Maximum Daily Use of Water by Month in Two NC Counties, 2020",
       color = "County") +
  theme(legend.position = "bottom")
```



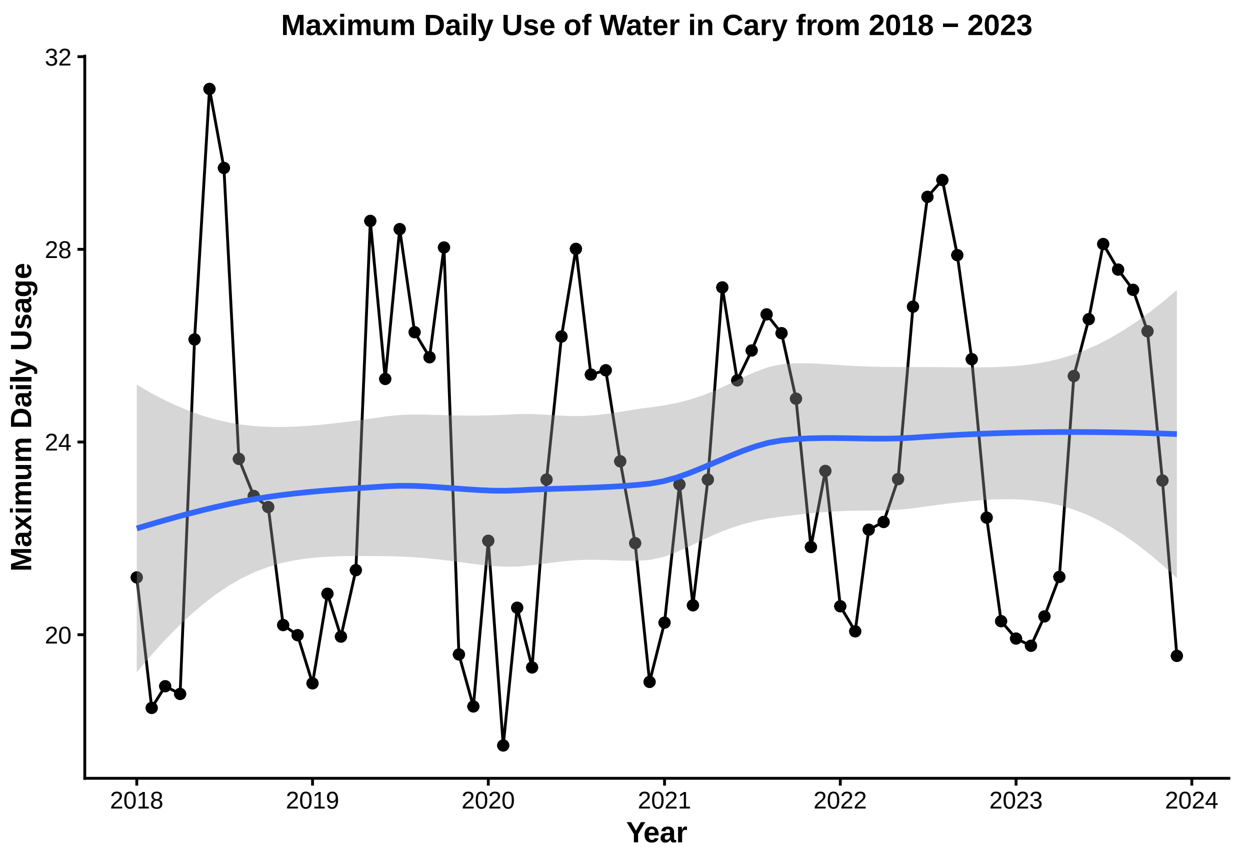
9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

   TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to `bind_rows()` to combine the dataframes into a single one, and use that to construct your plot.

```r
#9

# Create a vector of years 2018 - 2023
cary_mgd_years <- as.numeric(c(2018:2023))

# Get the data
cary_mgd <- map2('03-92-020', cary_mgd_years, lwsp_function) %>%
  bind_rows()

# Create the plot
ggplot(cary_mgd, aes(x = date, y = mgd_monthly_fcn)) +
  geom_point() +
```

```r
  geom_line() +
  geom_smooth(method = "loess") +
  labs(x = "Year", y = "Maximum Daily Usage",
       title = "Maximum Daily Use of Water in Cary from 2018 - 2023") +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Yes, it appears that water usage peaks in the middle of the year during the summer. There may be a slight increase in usage over the years, but there is not a definitive increasing trend here. >